



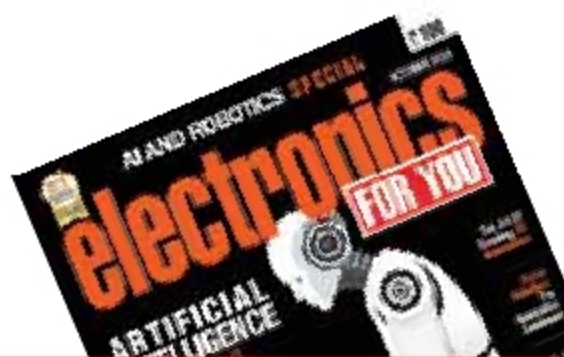
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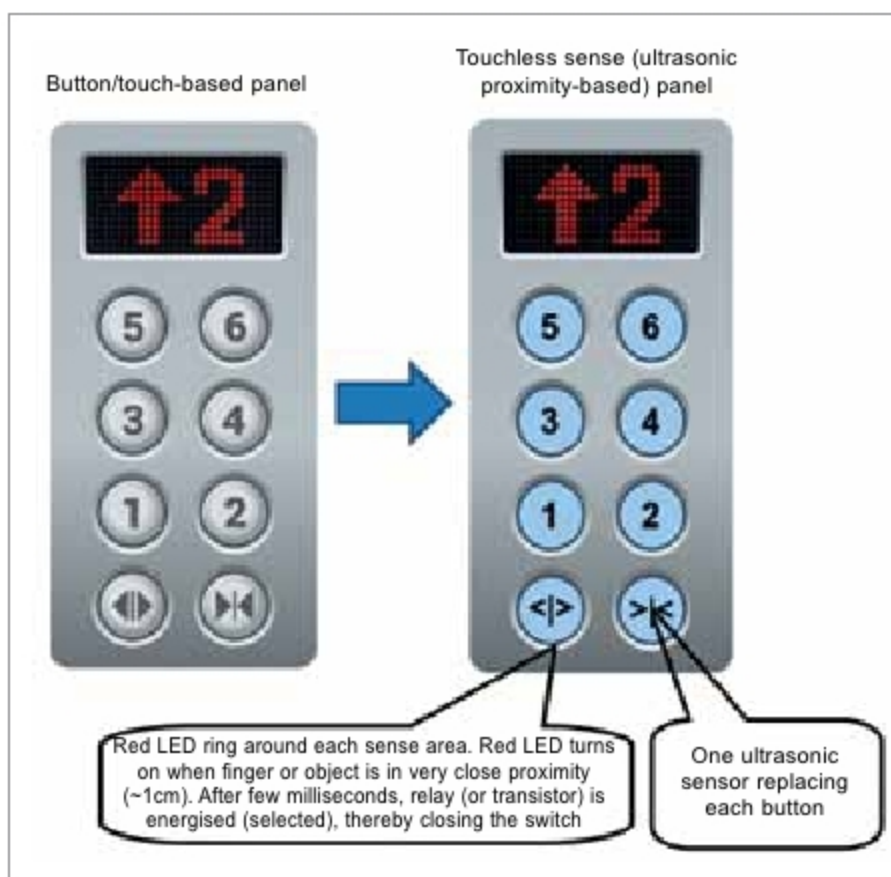


Fig. 3: Transformation from present button-based panel to touchless sense switching panel using ultrasonic sensors

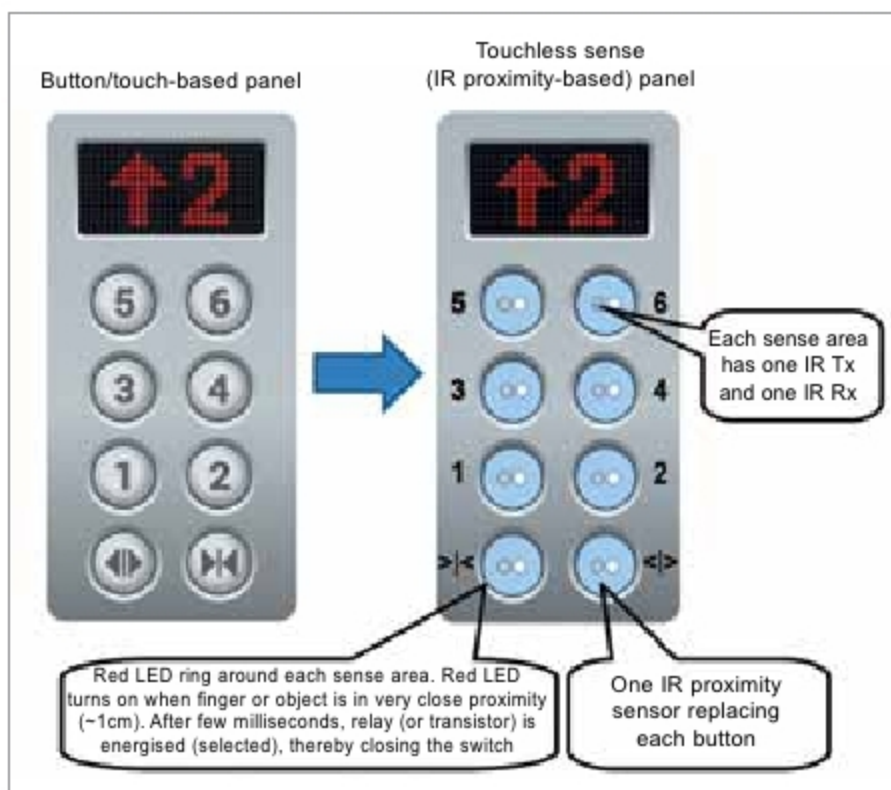


Fig. 4: Transformation from present button-based panel to touchless sense switching panel using infrared proximity sensors

Proposed event-driven software architecture of touchless sense switching is shown in Fig. 2.

Operating new touchless sense switching panels

Below are the two technologies that can be leveraged to realise the touchless sense switch panel as a solution.

Using ultrasonic proximity sensors. Fig. 3 shows the transformation from present button-based panel to touchless sense switching

panel using ultrasonic sensors. Each ultrasonic proximity sensor acts as an ultrasonic emitter and receiver.

Using infrared (IR) proximity sensors. Fig. 4 shows the transformation from present button-based panel to touchless sense switching panel using infrared proximity sensors. Each infrared proximity sensor has one IR transmitter and one IR receiver. When these two are put together, they act as one single sense area (next-generation switch).

Conclusion

TSS technology is a solution to address one of the problems caused due to the current pandemic. The existing physical human-machine interactions via touch panels and switches pose life threat during pandemic situations as viruses can spread through physical

contacts by infected/uninfected individuals.

In addition, the performance is maximised and wear-tear cost is reduced with the use of TSS. The design of TSS also gives the flexibility of configuring the behaviour of switching.

The functionality can be further extended to support virtual assistants like Alexa and Google Home with little hardware additions and make the equipment or device smarter. **EFY**